Lec 2: Factors Affecting bacterial growth

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Microbial biomass

refers to the total mass of microorganisms in a given environment, including bacteria, fungi, archaea, and protozoa. It is a measure of the living component of the microbial community and is often expressed as grams of biomass per unit area or volume.

Microbial biomass consists primarily of living cells, but it can also include non-living components, such as dead cells and extracellular materials produced by microorganisms.

Microbial biomass is often estimated using methods like direct cell counting (e.g., microscopy), biochemical assays (e.g., measuring ATP), or indirect methods (e.g., carbon or nitrogen content).



Industrial Microbiology

Control of growth is important for

- infection control
- growth of industrial and biotech
- organisms



Environmental factors

Nutritional Requirements

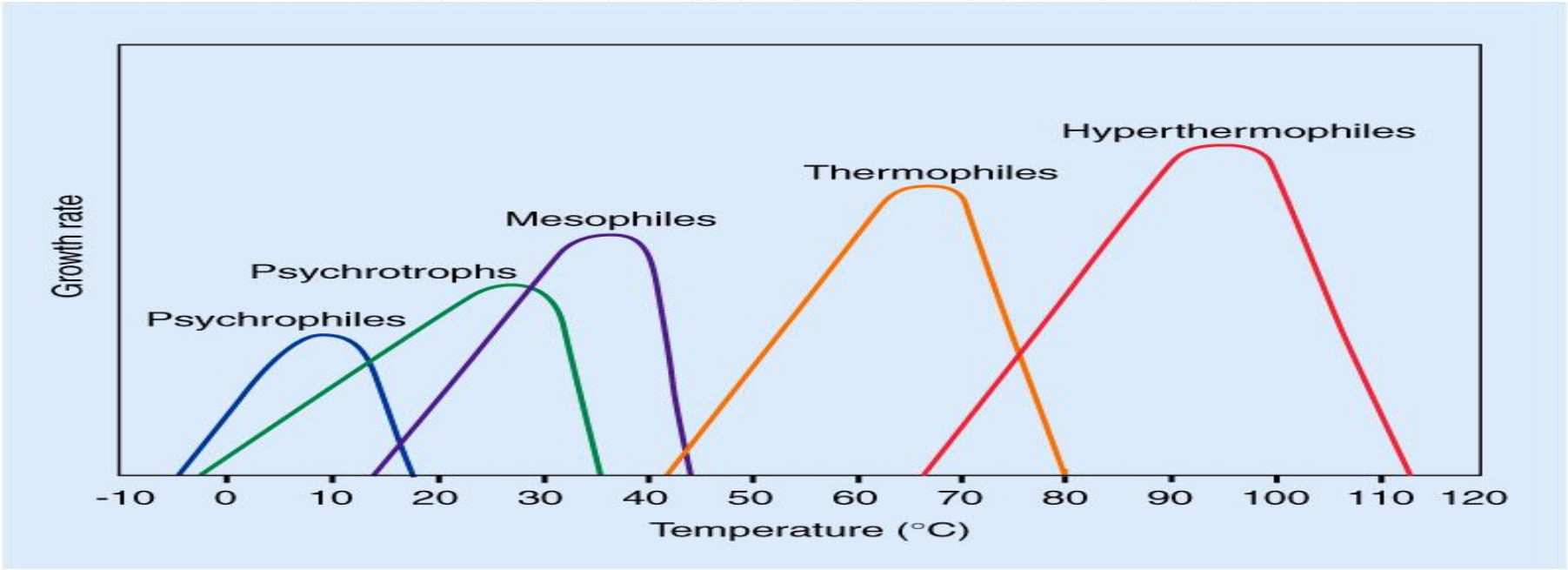
Growth Factors

Temperature

- **Enzymes**, the machinery of the cell, are influenced by external factors and can be shown to have a range where they function that includes an optimal value that produces the highest activity.
- The range of enzyme activity determines the range for growth of specific bacteria, analogously leading to a value for optimal growth rate.

Temperature

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The effect of temperature is greater in the case of the presence of water(**Autclave 121C/ 15-20min)**, because easily to penetrate and faster to damage cell proteins in the case of moist heat than in the case of dry heat (**oven 160-180 C/ 30-60 min**).

Temperature

1. **Psychrophilic** : It has the ability to grow well below the freezing point, like some food spoilage fungi. Its optimum temperature is about (15°C) and its range is from zero to 15°C.

Micrococcus, Bacillus, Staphylococcus, Pseudomonas,

Flavobacterium, and Pseudomonas fluorescens

2. **Thermophilic** : their optimum temperature is around 50°C. beside they have a high G + C content in their DNA

(Bacillus stearothermophilus)

3. **Mesophilic** : It is the largest group in the world of fungi. It has a temperature range of (15-35°C) .

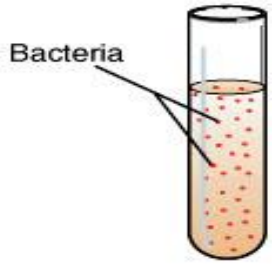
Lactobacillus, Lactococcus, Enterococcus, Pediococcus, Escherichia coli

Optimum temperature is ideal for growth as it does not affect the activity of enzymes and also does not affect the cellular protein.

Oxygen requirements

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Table 4.2 Oxygen (O_2) Requirements of Prokaryotes

	Obligate aerobe	Facultative anaerobe	Obligate anaerobe	Microaerophile	Aerotolerant
					
Enzymes in Cells for O_2 Detoxification					
Catalase: $2H_2O_2 \rightarrow 2H_2O + O_2$ Superoxide dismutase: $2O_2^- + 2H^+ \rightarrow O_2 + H_2O_2$		Catalase, superoxide dismutase	Neither catalase nor superoxide dismutase in most	Small amounts of catalase and superoxide dismutase	Superoxide dismutase

Bacteria are classified in terms of their need for oxygen for energy production into four main groups:

1. **Obligatory aerobes**: these need the presence of oxygen in a sufficient concentration for their growth ***Pseudomonas aeruginosa***
2. **Obligatory anaerobes**: they live and are active under anaerobic conditions. ***Clostridium tetani***
3. **Facultative anaerobes**: This group lives in the presence or absence of oxygen. ***Escherichia coli***
4. **Microaerophilic:** Organisms that require a small amount of oxygen. ***Azosprillium spp.***

The effect of rays:

The effect of radiation on microbes depends on the wavelength, intensity and length of exposure.

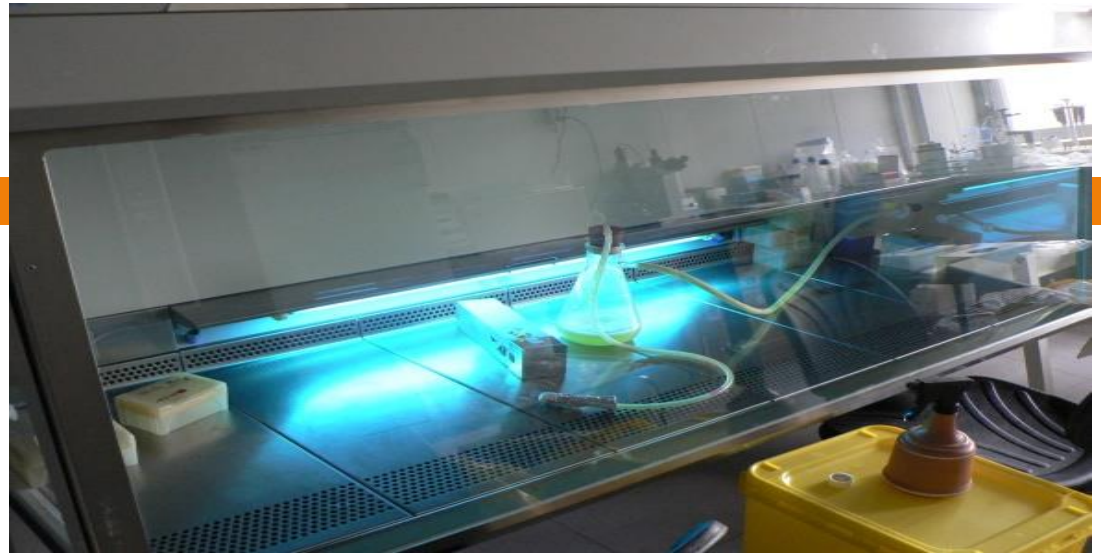
Two types of radiation are used for this purpose:

1. **Ionizing radiation**(harmful) : Like x-rays and gamma rays (harmful)
2. **Non ionizing radiation** (harmful): They are rays with shorter wavelengths than those that characterize ionizing radiation,

for example, **ultraviolet light radiation**, which explains the use of ultraviolet light bulbs in sterilization through its effect on cell proteins and DNA, as it works to form free groups and produce hydrogen peroxide.

- Ultraviolet rays are part of the visible spectrum with wavelengths ranging between (204-260) nanometers, which are suitable for producing **mutations** and are less dangerous than X-rays, which have the ability to cause mutations in microbial cells, leading to cell death.

Ultraviolet rays



- Ultraviolet rays are often used now in laboratories, hospitals, drug and some food factories to kill the microbes present in the air of rooms or deposited on the surfaces of various materials in the room.

- For this purpose, quartz mercury vapor bulbs that give off ultraviolet rays of a wavelength around 2650 angstroms are used for this purpose.
- It is the longest of these rays that kill microbes, and many of these lamps are hung in suitable places on the ceilings or on the walls, or placed in tubes for the air entering the room to pass through and upon exposure to the wavelength of 2650 angstroms, it kills in seconds depending on the type of pollution, its extent, the size of the vacuum, the humidity and temperature of the atmosphere and the type of radiation And the time of exposure. When using ultraviolet rays, workers take into account not to be exposed to it for more than several minutes a day in order to avoid the occurrence of facial burns or eye damage.

pH

Bacterial growth is affected by changes in the hydrogen ion concentration in the growing media.

Microbial growth stops at high acidity and alkalinity, which affects enzymatic activity and metabolic processes.

The degree of hydrogen ion concentration appropriate for growth is not necessarily appropriate for biological processes, production of enzymes, pigments,

- Neutrophiles grow best around neutral pH (7)
- Acidophiles grow best at pH < 7
- Alkophiles grow best at pH > 7
- Acidotolerant grow best at pH 7 but can also grow at lower pH
- Alkotolerant grow best at pH 7 but can also grow at higher pH

Water availability

- Liquid water is essential for life.
- Aqueous solutions actually have different amounts of water available, depending on how many solutes are dissolved in it. As a very simple model, consider two glasses, one full of pure water, the other containing the same amount of water plus a sponge. Which one would be easier to drink? On a much smaller scale, dissolved solutes act like a sponge, making less water available.
- Water activity (a_w) can be decreased by the addition of any soluble molecule although salt (NaCl) and sugars are probably the most common.

NaCl Tolerance

Osmosis is the net movement of solvent molecules through a partially permeable membrane into a region of higher solute concentration**. The intent of osmosis is to equalize the solute concentrations on the two sides of the cells.**

- **Non-halophiles:** Microbes that require a high water activity require less than 1% salt (***Escherichia coli*** or ***Pseudomonas***)
- moderate halophiles require 6-15% salt (***Staphylococcus aureus***)
- Some bacteria require salt to grow and are called **halophiles**. If a very high concentration of salt is required (around saturation), the organisms are termed **extreme halophiles**. Archaea such as ***Halococcus***
- **Halotolerant-** bacteria that are able to grow at moderate salt concentrations, even though they grow best in the absence of NaCl

Nutritional Requirements

- Growth of prokaryotes depends on nutritional factors as well as physical environment
- Main factors to be considered are:
  - Required elements
  - Growth factors
  - Energy sources

Nutritional Requirements

- Major elements (C-H-O-N-P-S + K, Mg, Fe, Ca)
  - Carbon, oxygen, hydrogen, nitrogen, sulfur, phosphorus, potassium, magnesium, iron, and calcium
    - Essential components for macromolecules
- Organisms classified based on carbon usage
  - Heterotrophs
    - Use organism carbon as nutrient source
  - Autotrophs
    - Use inorganic carbon (CO_2) as carbon source
- Trace elements (Co, Cu, Ni, Zn, Se, Mg, Wo)
  - Cobalt, zinc, copper, molybdenum and manganese
    - Required in minute amounts
    - Assist in enzyme function
- Nutritional diversity
  - Different organisms require the same nutrients but may require different forms of the nutrients

Major elements

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Table 4.3 Representative Functions of the Major Elements

Chemical	Function
Carbon, oxygen, and hydrogen	Component of cellular constituents including amino acids, lipids, nucleic acids, and sugars.
Nitrogen	Component of amino acids and nucleic acids.
Sulfur	Component of some amino acids.
Phosphorus	Component of nucleic acids, membrane lipids, and ATP.
Potassium, magnesium, and calcium	Required for the functioning of certain enzymes; additional functions as well.
Iron	Part of certain enzymes.

Carbon Source

- Organic molecules
  - Heterotrophs
- Inorganic carbon (CO_2)
  - Autotrophs

Nitrogen Source

- Organic nitrogen
  - Primarily from the catabolism of amino acids
- Oxidized forms of inorganic nitrogen
  - Nitrate (NO_3^{2-}) and nitrite (NO_2^-)
- Reduced inorganic nitrogen
  - Ammonium (NH_4^+)
- Dissolved nitrogen gas (N_2) (Nitrogen fixation)

Phosphate Source

- Organic phosphate
- Inorganic phosphate ($H_2PO_4^-$ and HPO_4^{2-})

Sulfur Source

- Organic sulfur
- Oxidized inorganic sulfur
  - Sulfate (SO_4^{2-})
- Reduced inorganic sulfur
  - Sulfide (S^{2-} or H_2S)
- Elemental sulfur (S_o)

Growth Factors

- Typical molecules
  - Amino acids
  - Nucleotide bases
  - Enzymatic cofactors or “vitamins”